

EX. NO: 5

IMPLEMENTATION OF RAIL FENCE (ROW & COLUMN) TRANSFORMATION TECHNIQUE

AIM

To write a C program to implement the Rail Fence Transposition Cipher encryption technique.

ALGORITHM

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the number of rails (rows).

Step 4: Create a matrix with the given number of rails and columns equal to the length of the plaintext.

Step 5: Write the plaintext characters diagonally in a zigzag pattern across the rails.

Step 6: After filling the matrix, read the characters row by row to generate the ciphertext.

Step 7: Display the Rail Fence matrix and the ciphertext.

Step 8: Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char text[100], rail[10][100];
```

```
    int i, j, len, key;
```

```
    printf("Enter Plain Text: ");
```

```
    scanf("%s", text);
```

```
    printf("Enter Number of Rails: ");
```

```
    scanf("%d", &key);
```

```
    len = strlen(text);
```

```
    // Fill matrix with '*'
```

```
    for(i = 0; i < key; i++)
```

```
        for(j = 0; j < len; j++)
```

```
            rail[i][j] = '*';
```

```

int row = 0, dir = 1;
// Fill Rail Fence matrix
for(i = 0; i < len; i++)
{
    rail[row][i] = text[i];
    if(row == 0)
        dir = 1;
    else if(row == key - 1)
        dir = -1;
    row += dir;
}
printf("\nRail Fence Matrix:\n");
for(i = 0; i < key; i++)
{
    for(j = 0; j < len; j++)
        printf("%c ", rail[i][j]);
    printf("\n");
}
printf("\nCipher Text: ");
for(i = 0; i < key; i++)
{
    for(j = 0; j < len; j++)
    {
        if(rail[i][j] != '*')
            printf("%c", rail[i][j]);
    }
}
printf("\n");
return 0;
}

```

OUTPUT:

The screenshot displays the Programiz C Online Compiler interface. The browser's address bar shows the URL `programiz.com/c-programming/online-compiler/`. The page header includes the Programiz logo, a banner for Adelaide University with the text "LAUNCH INTO THE TECH WORLD", and a "Programiz PRO" button.

The main workspace is divided into two panels. The left panel, titled "main.c", contains the following C code:

```
36
37 printf("\nRail Fence Matrix:\n");
38 for(i = 0; i < key; i++)
39 {
40     for(j = 0; j < len; j++)
41         printf("%c ", rail[i][j]);
42     printf("\n");
43 }
44
45 printf("\nCipher Text: ");
46
47 for(i = 0; i < key; i++)
48 {
49     for(j = 0; j < len; j++)
50     {
51         if(rail[i][j] != '*')
52             printf("%c", rail[i][j]);
53     }
54 }
55
56 printf("\n");
57
58 return 0;
59 }
```

The right panel, titled "Output", shows the program's execution results:

```
Enter Plain Text: pallaviIsAGoodGirl
Enter Number of Rails: 3

Rail Fence Matrix:
p * * * a * * * s * * * o * * * r *
* a * l * v * I * A * o * d * i * l
* * l * * i * * * G * * * G * * *

Cipher Text: pasoralvIAodilliGG

=== Code Execution Successful ===
```

The bottom of the image shows a Windows taskbar with the date and time set to 09:10 PM on 03-08-2026, and the weather as 27°C Mostly cloudy.